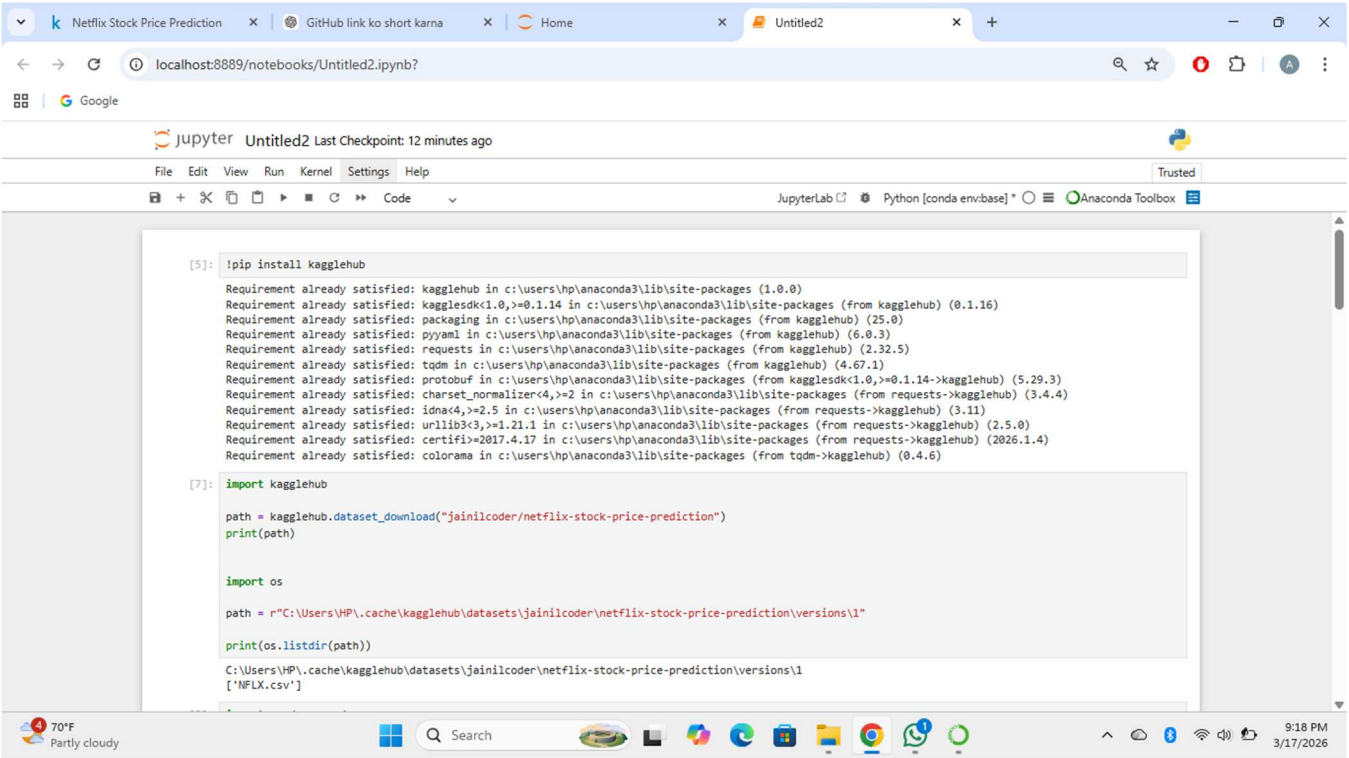
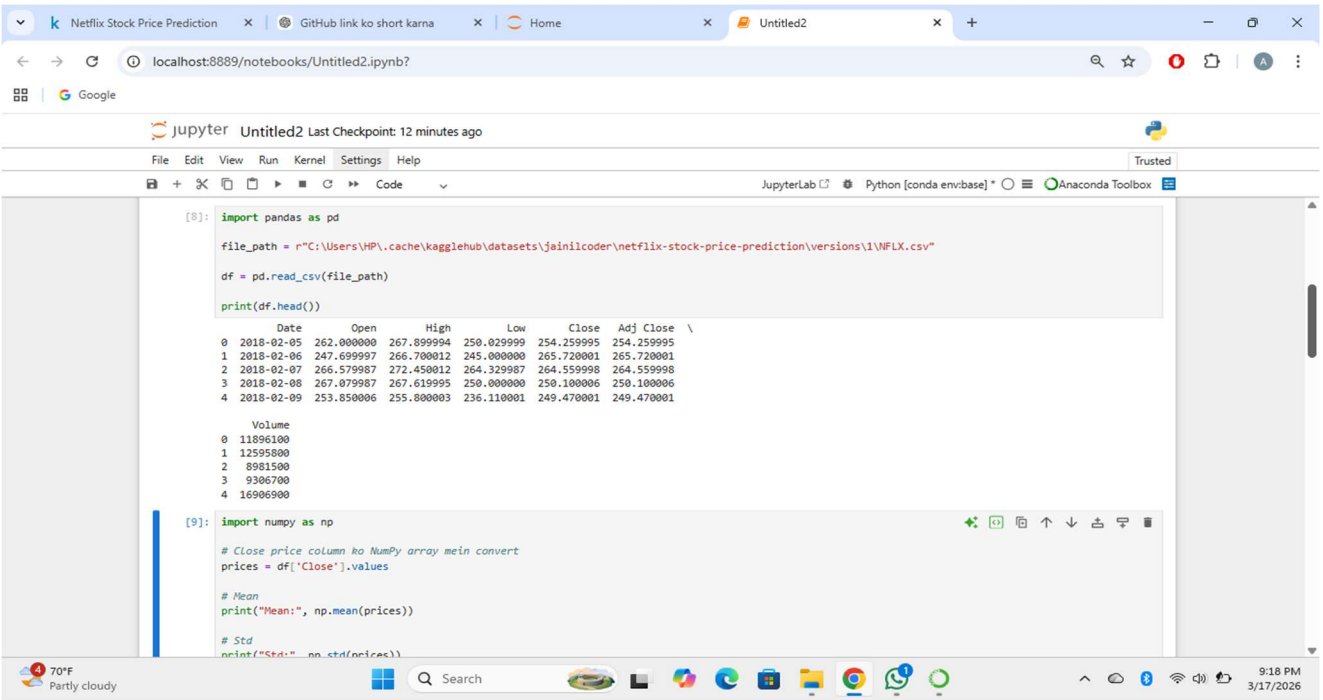
Install libraries:



Create dataset:



Array:

```
*[10]: #array
arr1 = np.array(prices[:5])
arr2 = arr1.reshape(5,1)
arr3 = arr2.reshape(5,1,1)

print("\n1D Array:", arr1)
print("\n2D Array:\n", arr2)
print("\n3D Array:\n", arr3)
```

```
1D Array: [254.259995 265.720001 264.559998 250.100006 249.470001]

2D Array:
 [[254.259995]
 [265.720001]
 [264.559998]
 [250.100006]
 [249.470001]]

3D Array:
 [[[254.259995]]

 [[265.720001]]

 [[264.559998]]

 [[250.100006]]

 [[249.470001]]]
```

```
*[11]: #Indexing & Slicing
print("\nFirst element:", arr1[0])
```

Indexing, slicing, reshaping, operations:

```
*[11]: #Indexing & Slicing
print("\nFirst element:", arr1[0])
print("Last 3 elements:", arr1[-3:])
print("All rows second column (2D):", arr2[:,0])
```

```
First element: 254.259995
Last 3 elements: [264.559998 250.100006 249.470001]
All rows second column (2D): [254.259995 265.720001 264.559998 250.100006 249.470001]
```

```
*[12]: #reshaping
reshaped = prices[:10].reshape(5,2)
print("\nReshaped (5x2):\n", reshaped)
```

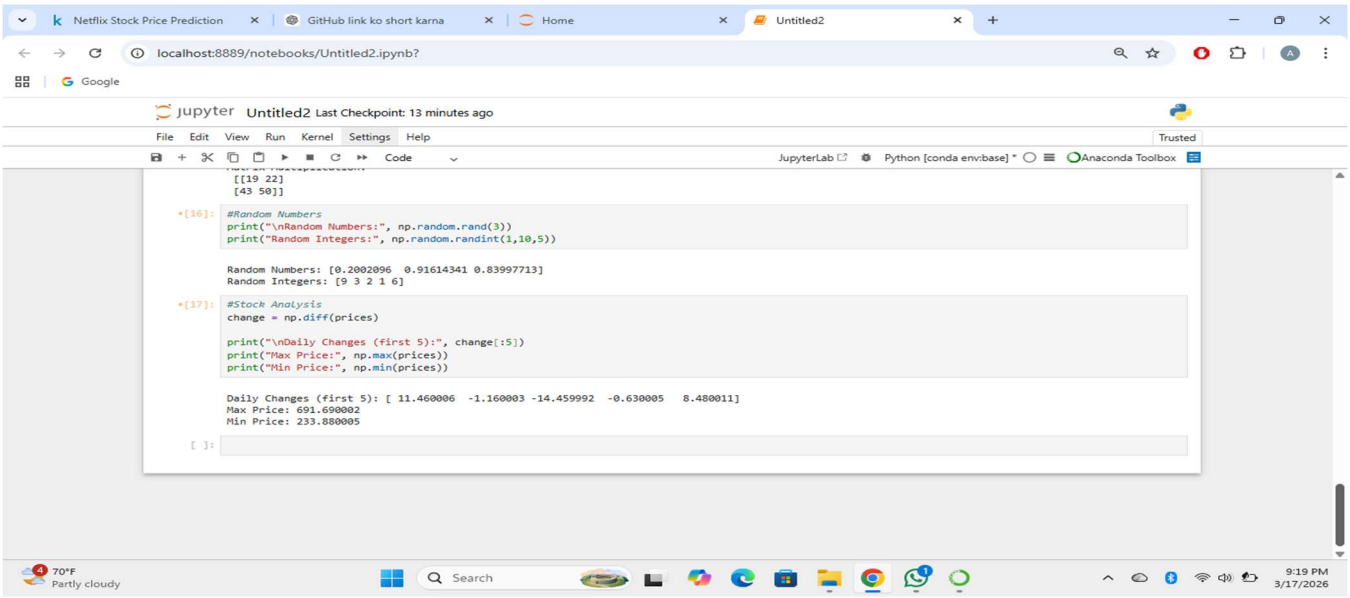
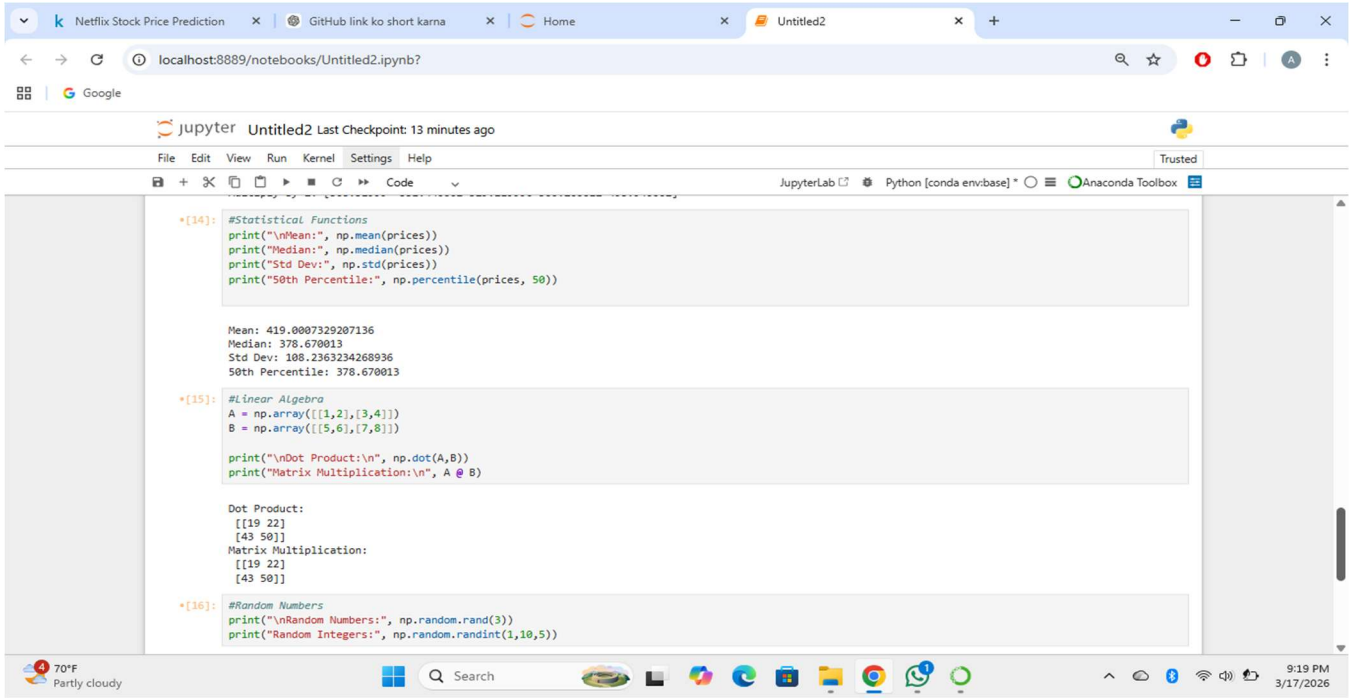
```
Reshaped (5x2):
 [[254.259995 265.720001]
 [264.559998 250.100006]
 [249.470001 257.950012]
 [258.269989 266.      ]
 [280.269989 278.519989]]
```

```
*[13]: #Mathematical Operations
print("\nAdd 10:", arr1 + 10)
print("Multiply by 2:", arr1 * 2)
```

```
Add 10: [264.259995 275.720001 274.559998 260.100006 259.470001]
Multiply by 2: [508.51999  531.440002 529.119996 500.200012 498.940002]
```

```
*[14]: #Statistical Functions
```

Statistical functions and linear algebra:



Random number: